Ramkrishna Acharya

Machine Learning Researcher | M.Sc. in Data Science at FAU
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Education

Katholieke Universiteit Leuven (KU Leuven) — *Leuven, Belgium*
PhD. Bioscience Engineering — *2026 – Present*

- **Topic:** *Context-aware insect monitoring in agriculture using AI.* | Profile
- **Promoter:** Prof. Wouter Saeys

Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU) — *Erlangen, Germany*
M.Sc. Data Science, German grade scale: 1.8 (Good) — *2022 – 2025*

- **Thesis:** *Enhancing Smart Grid Security: A Deep Learning Approach to Adversarial Intrusion Detection* | Code | PDF
- **Supervisors:** Dr.-Ing. Loui Al Sardy, Mamdouh Muhammad, M. Sc.
- **Focus:** Applied machine learning, deep learning, and cyber-physical security

Tribhuvan University — *Kathmandu, Nepal*
B.Sc. Computer Science and Information Technology, grade: 72.07/100 (First Division) — *2015 – 2019*

- **Research Project:** *Devanagari Handwritten Character Recognition* | Code
- **Focus:** Image processing and neural networks

Experience

Chair for Computer Science 7, FAU Erlangen-Nürnberg — *Erlangen, Germany*
Master's Thesis Researcher and HiWi — *Apr 2025 – Mar 2026*

- Developed *ROSAID-ST* and *ADVIS-G*, an adversarially defended intrusion-detection system for smart-grid communication data.
- Prepared exercise materials and tutored students in the course *Cybersecurity Fundamentals in Networking.*

Primetals Technologies Germany GmbH — *Erlangen, Germany*
Working Student – Computer Vision — *May 2023 – Mar 2026*

- Engineered industrial vision pipelines for detection, segmentation, tracking, and anomaly analysis.
- Built synthetic data and annotation tools for OCR and segmentation under limited-data settings.
- Developed FastAPI/SQL-based model result evaluation workflows, saving 100+ hours of manual video analysis.
- Benchmarked, trained, and fine-tuned novel and SOTA OCR, detection, tracking, anomaly, video, and time-series models.

Extensodata Pvt. Ltd. — *Kathmandu, Nepal*
Associate Data Scientist — *Aug 2021–Aug 2022*

- Led a team of 3 and developed a business intelligence tool using Python, MySQL, Pentaho, Apache Airflow, and Apache Superset.
- Developed fintech data pipelines and ML workflows to support loan-risk analysis.

Upwork.com — *Remote*
Freelancer (Profile) — *Active*

- *Jan 2022 – Aug 2022:* Built a real-time, dynamic alert-based strategy backtesting platform for options and stock trading.
- *Mar 2026 – Present:* Delivering full-stack AI and computer-vision applications for freelance clients.

Additional Experience

- **diyo.ai**, Unity3D Development Intern, Kathmandu, Nepal — Nov 2020 – Mar 2021
- **MPercept Technology**, AI Developer Intern, Kathmandu, Nepal — Aug 2019 – Jan 2020

Publications

[1] **Ramkrishna Acharya**, Loui Al Sardy, Mamdouh Muhammad, Reinhard German
ADVIS-G: An Adversarially Defended Intrusion Detection System for Smart Grids Using Deep Learning
KI – Künstliche Intelligenz, 2026. DOI | Code
Proposed a framework of an adversarially robust deep learning intrusion detection system using image-based traffic representation.

[2] **Ramkrishna Acharya**, Loui Al Sardy, Mamdouh Muhammad
ROSAID-ST: Adversarially Robust Intrusion and Anomaly Detection in ROS Networks via Student-Teacher Deep Learning
International Journal of Advanced Robotic Systems, under review, 2026. Journal | Code
Developed a ROS network IDS combining CNN-based attack detection with student-teacher anomaly detection.

Recent Projects

- **Session Feature Extractor** (2026): PyPI tool for extracting image features from network traffic data.
- **Image Baker** (2025): PyPI tool for annotation and synthetic data generation for anomaly detection.
- **SmokeSim** (2024): PyPI package for smoke simulation, image augmentation, and segmentation.
- **Neural Template Matching** (2024): custom U-Net experiment for template matching on the COCO dataset.

Skills

Programming	Python (advanced), SQL, Bash, JavaScript (basic), C# (basic)
Machine Learning	PyTorch, TensorFlow, Scikit-learn, HuggingFace, MMEngine, OpenCV
ML Systems	Docker, Git/GitHub, MLflow, DVC, ONNX, TensorRT, Apache Airflow, FastAPI, Flask, Streamlit, Firestore, AWS EC2, GitHub Actions, GitLab CI/CD, Apache Superset

Languages

English (C1), German (A2), Nepali (Native)